

Quantum Mechanics in Solids

Unit-5

Lecture-1

Free electron theory of metals

Free Electron Theory of Metals

Cu

Fe, Ni, Co

Si & Ge

Glass → ●

Electrical,
Optical
Magnetic
Thermal...

ELECTRON THEORY OF SOLIDS

INTRODUCTION

The electron theory of solids aims to explain the structures and properties of solids through their electronic structure.

- ❖ Electrical conduction in metals.

- ❖ Ohms law.

- ❖ Thermal properties.

- ❖ Specific heat

- ❖ Optical properties

- ❖ Magnetic properties

Free electron theory of solids

The electron theory of solids has been developed in three main stages.

✓ (i) The classical free electron theory:

{ Drude and Lorentz developed this theory in 1900. According to this theory, the metals containing free electrons obey the laws of classical mechanics.

✓ (ii) The Quantum free electron theory:

Somerfield developed this theory during 1928. According to this theory, the free electrons obey quantum laws.

✓ (iii) The Zone Theory:

Bloch stated this theory in 1928. According to this theory, the free electrons move in a periodic field provided by the lattice. This theory is also called "Band theory of solids".

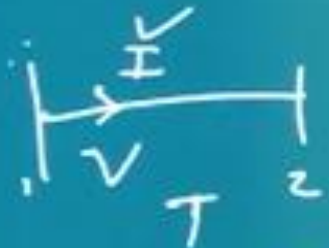
Free electron theory of solids

❖ Electric current

The rate of flow of charge through a conductor is known as the current.

If a charge ' dq ' flows through the conductor for ' dt ' second, then

$$\text{Electric current } (I) = \frac{dq}{dt}$$



❖ Ohm's law

At (constant temperature), the potential difference between the two ends of a conductor is directly proportional to the current that passing through it.

$$V \propto I \text{ (or) } V = IR$$

where R = resistance of the conductor

❖ Resistance of a conductor

The resistance (R) of a conductor is the ratio of the potential difference (V) applied to the conductor to the current (I) that passes through it.

$$\text{Resistance } (R) = \frac{V}{I}$$

❖ The specific resistance (or) resistivity of a conductor:

The resistance (R) of conductor depends upon its length (L) and cross-sectional area (A) i.e.,

$$R \propto \frac{L}{A} \quad \text{or} \quad R = \rho \frac{L}{A}$$

where ' ρ ' is a proportional constant and is known as the specific resistance (or) resistivity of the material.

The **resistivity** of a material is defined as the resistance of the material having unit length and unit cross sectional area.

$$\rho = R \frac{A}{L} \text{ ohm metre}$$



$$\rho = R$$

Units of ρ : Ωm

The electrical conductivity (σ) of a conductor

The reciprocal of the electrical resistivity is known as electrical conductivity (σ) and is expressed in $\text{ohm}^{-1} \text{ metre}^{-1}$.

The conductivity (σ)

$$\sigma = \frac{1}{\rho} = \frac{1}{\left(\frac{RA}{L}\right)} = \frac{L}{RA} \quad \text{ohm}^{-1} \text{ metre}^{-1}.$$

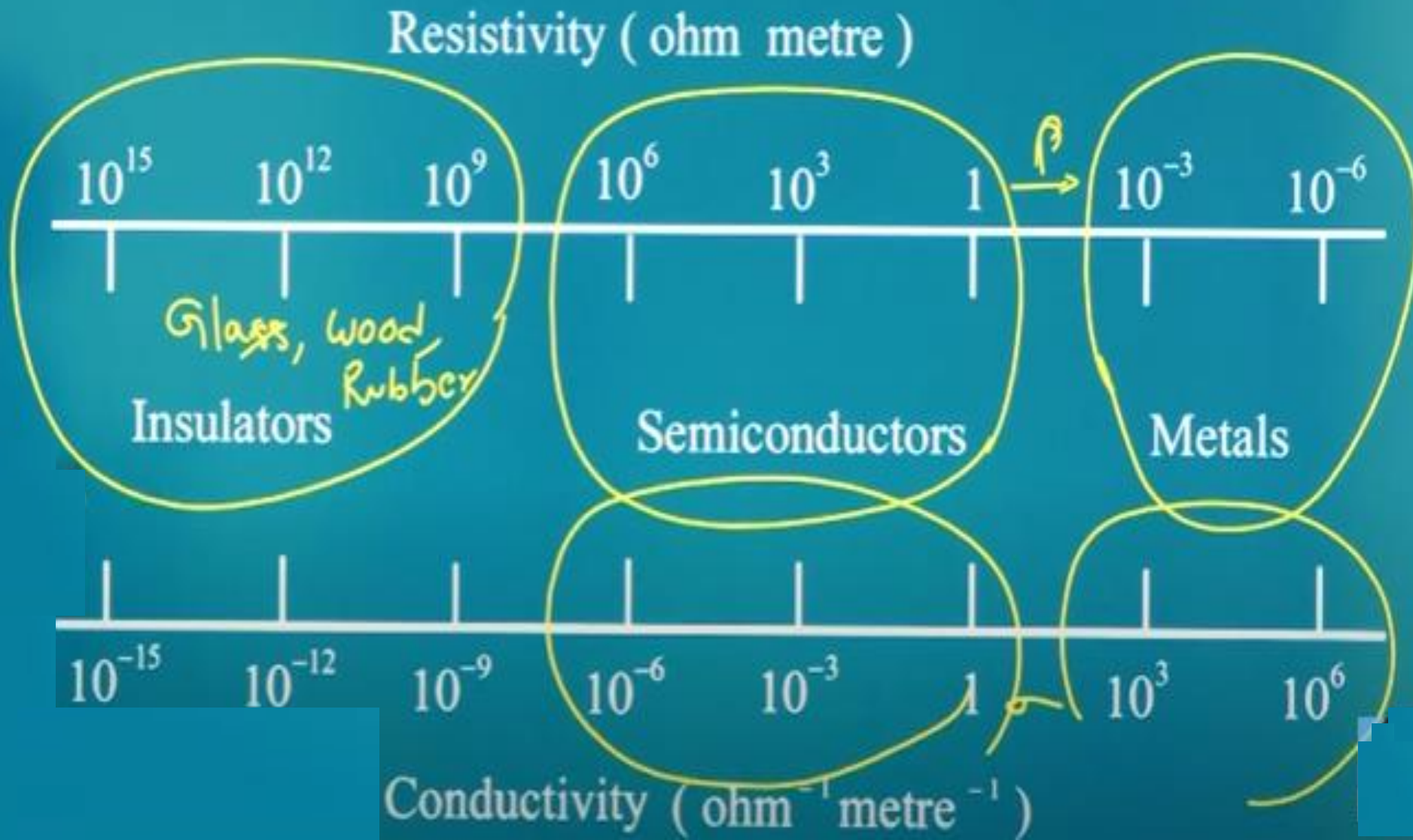
Free electron theory of solids

In terms of electrical properties, the materials can be divided into three groups

- ✓ ❖ Conductors → high electrical conductivity (or) low ρ
- ✓ ❖ Semi conductors and → Intermediate b/w conductors & insulators
- ✓ ❖ Dielectrics (or) insulators. → high electrical resistivity

Free electron theory of solids

Conductivities and resistivities of materials



* Drude - Lorentz Theory

OR

Free Electron Gas Model

OR

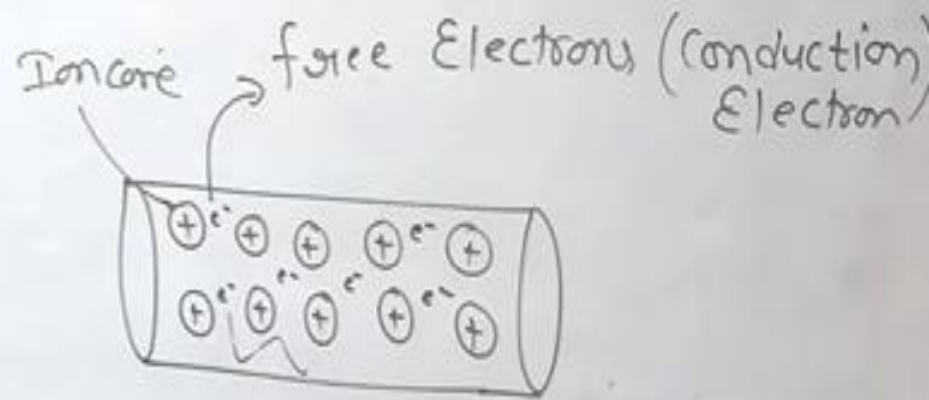
Classical Theory of Free Electron:-

Assumption:

(1) There is no interaction b/w Ion cores (+ve) and free Electrons.

(2) Free electrons are randomly move in all direction, which are bound in certain volume.

(3) Total Energy of free Electron in metal is whole kinetic $(P.E)_{\text{free electron}} = 0$ in Metal



(4) Free Electrons are considered as the atoms/molecules of perfect Gas

Free Electron Gas

Drude's model

e^- s were classical particles
and the e^- s were like a gas of
particles enclosed in a box
↓
metal

e^- gas : Assumption.



- ① ' τ ': Mean time interval
between 2 successive collisions
experienced by an e^- is ' τ '.

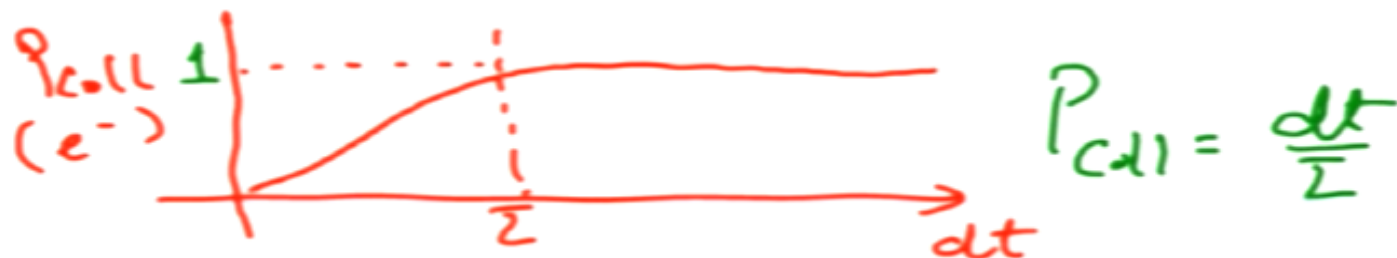


Probability for an e^- in this gas to undergo a collision in time dt is

$$P_{\text{coll}} = \frac{dt}{\tau} \quad dt \leq \tau$$

$\tau : 10^{-14}$ sec ; $P_{\text{coll}} = 1, dt > \tau$

Drude had considered was that the collision occur between the e^- & ion cores



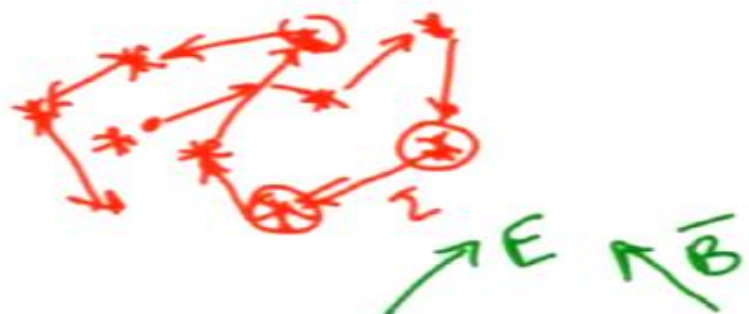
- ② Between two successive collisions of an e^- , the motion of the particle can be affected by $\vec{E}$ & $\vec{B}$

$\vec{E}$: electric field

$\vec{B}$: Magnetic field.

$$\langle p \rangle = 0$$

$\uparrow$
 $dt \gg \tau$



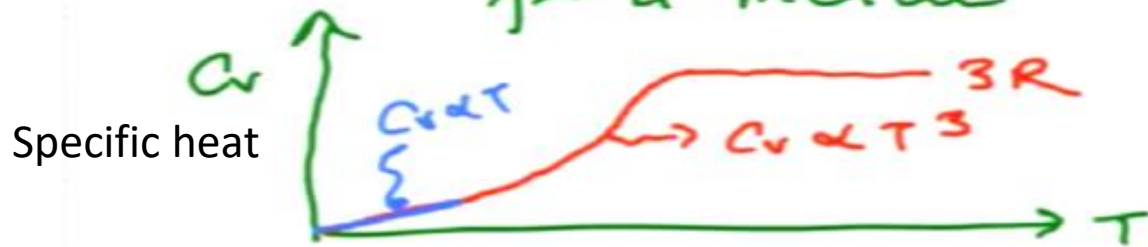
Collisions have completely randomized the momentum of the e^- .

$\langle p \rangle = 0$ ∴ Kinetic theory of gases

Between two successive collision the $\vec{P}$,



- ③ e^- in the electron gas are free electron,
 Non interacting electrons
 Free electron approximation for a metal



Advantages of classical free electron theory

- 1) It explains the electrical conductivity and thermal conductivity of metals.
- 2) It explains the Wiedemann - Franz law.
- 3) It verifies Ohm's law.
- 4) It is used to explain the optical properties of metals.

Drawbacks of classical free electron theory

- 1) It fails to explain the electric specific heat and the specific heat capacity of metals.
- 2) It fails to explain superconducting properties of metals.
- 3) It fails to explain new phenomena like photo-electric effect, Compton effect, Black body radiation, etc.
- 4) It fails to explain electrical conductivity of semiconductors or insulators.

- 5) The classical free electron model predicts the incorrect temperature dependence of σ .
- 6) It fails to give a correct mathematical expression for thermal conductivity.
- 7) Ferromagnetism could not be explained by this theory.
- 8) Susceptibility has greater theoretical value than the experimental value.